

INFOSYS SPRINGBOARD PROJECT

Dashboard Storyboard & Architecture Blueprint

MedTrack Hospital Operations & Patient Analytics

A comprehensive technical specification, wireframe architecture, and interaction plan for enterprise healthcare analytics visualization.

DELIVERABLE MODULE

Module 4 – Planning & Prototyping

TARGET PLATFORM

Tableau Desktop / Server

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DOCUMENT VERSION

v1.0 (Enterprise Final)

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Executive Document Brief

This storyboard serves as the authoritative architectural blueprint for the **MedTrack Hospital Operations & Patient Analytics Dashboard** system. Designed for the Infosys Springboard curriculum under Module 4, this technical document details the operational workflow, metric engineering formulas, low-fidelity and mid-fidelity UI layouts, interactive parameters, and UI design standards required prior to Tableau production development.

Document Scope: To establish a standard for high-performance visual analytics across four key healthcare operational domains: High-level Executive Overview, Patient Throughput, Departmental Bottlenecks, and Medical Resource Allocation.

3. PROJECT OVERVIEW

3.1 Business Context & Objective

Modern healthcare organizations struggle with fragmented data streams across admission logs, departmental shift reports, bed management tools, and discharge registries. The primary objective of the **MedTrack Analytics Platform** is to unify these disparate operational datasets into an executive-grade dashboard suite built in Tableau.

3.2 Problem Statement

MedTrack Health Network faces operational inefficiencies, including:

- Unpredictable emergency department surge periods leading to high wait times.
- Sub-optimal bed turnaround times driving up the Average Length of Stay (ALOS).
- High 30-day readmission rates in specific clinical departments without clear visibility.
- Inflexible shift scheduling resulting in localized clinical staff burnout and equipment underutilization.

3.3 Project Scope

The project encompasses data cleaning, transformation, KPI engineering, wireframe designing, and building an interactive 4-view Tableau workbook (`medtrack\_prototype.twbx`).

3.4 Target User Personas

User Role	Primary Objective	Key Information Needs
Chief Medical Officer (CMO)	Executive oversight & strategic hospital performance	Network admission trends, 30-day readmissions, occupancy rates.
Operations Director	Throughput optimization & resource balancing	Patient flow bottlenecks, peak admission hours, bed allocation.
Department Heads	Clinical department performance & efficiency	Patient volume, treatment times, departmental efficiency scores.
Resource Managers	Physical asset & human capital planning	Bed utilization rates, ICU capacities, duty roster shifts, ventilator/MRI usage.

3.5 Expected Business Outcomes

Reduce average discharge turnaround time by **18%**, decrease unallocated bed time by **25%**, and provide leadership with daily actionable operational insights via automated visual reporting.

4. DASHBOARD ARCHITECTURE

The MedTrack BI architecture follows an end-to-end data pipeline flow from raw enterprise health records to final interactive decision support panels.

4.1 End-to-End Enterprise Data Flow



4.2 Architecture Technical Specifications

- **Data Extraction Strategy:** Daily scheduled extract refreshes (`.hyper`) configured to balance performance and update frequency.
- **Data Governance:** Column-level masking for Patient PII compliance (HIPAA alignment).
- **Performance Tuning:** Pre-aggregated Level of Detail (LOD) calculations to ensure visual query responsiveness under 2 seconds.

5. DASHBOARD PLANNING STRATEGY

The MedTrack system is modularized into four distinct dashboard views, tailored to different functional tiers within the hospital ecosystem.

Dashboard 1: Hospital Overview

Purpose: Executive oversight for executive management, offering macro-level organizational metrics.

Target Users: Chief Executive Officer, Chief Medical Officer, Operations Board.

Primary Insights: Network total admissions, overall occupancy rates, high-level readmission trends, and overall hospital efficiency status.

Dashboard 2: Patient Flow Analytics

Purpose: Analysis of patient movement, admission dynamics, and duration of stay patterns.

Target Users: Patient Intake Coordinators, ER Chiefs, Nursing Supervisors.

Primary Insights: Hourly peak arrival times, emergency vs. elective admission ratios, patient transfer lags, and ALOS trends.

Dashboard 3: Department Analytics

Purpose: Micro-level operational review across Cardiology, Orthopedics, ER, Pediatrics, and ICU.

Target Users: Clinical Department Chairs, Unit Head Nurses, Medical Audit Teams.

Primary Insights: Unit efficiency scores, treatment success metrics, unit-specific readmission percentages, and patient volume distribution.

Dashboard 4: Resource Utilization

Purpose: Optimization of high-value physical assets, bed inventory, medical equipment, and clinical workforce.

Target Users: Facilities Director, Nurse Staffing Planners, Equipment Procurement Teams.

Primary Insights: ICU/General Bed saturation, staff-to-patient ratios, equipment uptime, and localized capacity bottlenecks.

6. KPI PLANNING & FORMULATIONS

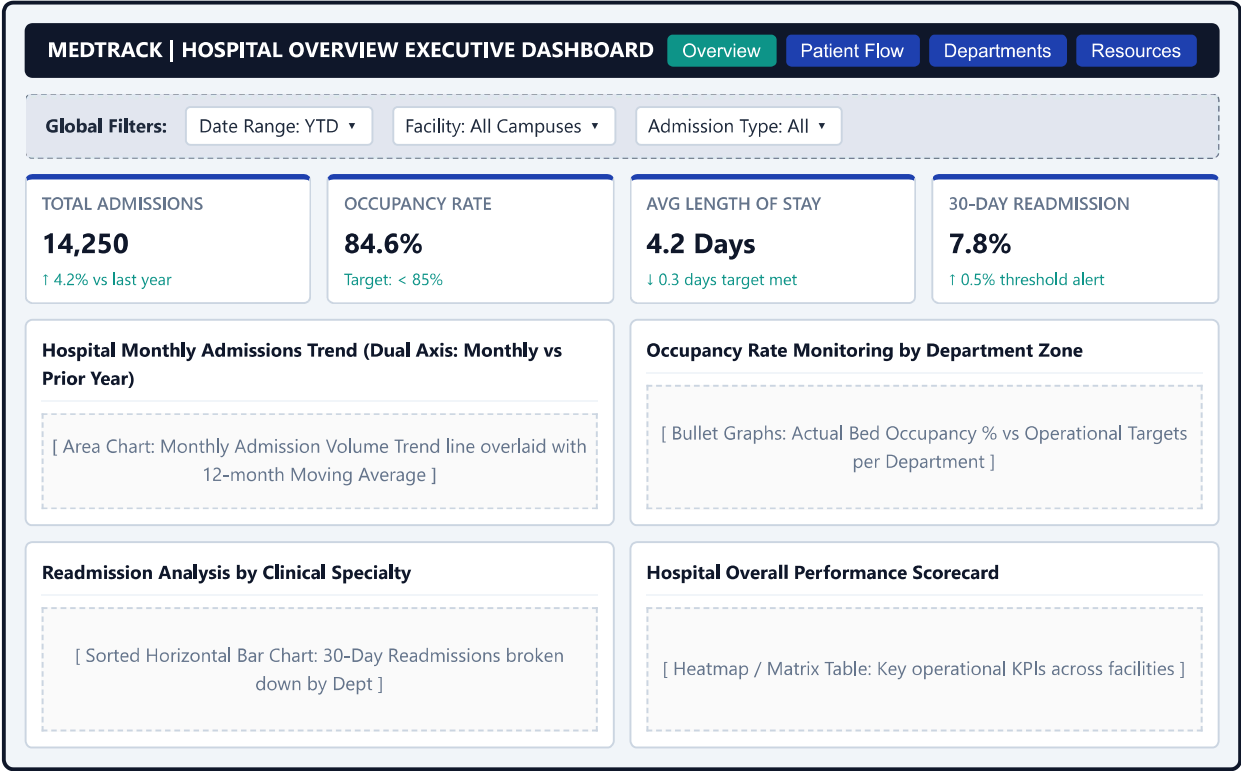
Below is the core visual metric framework defined for implementation in Tableau via Calculated Fields.

KPI Name	Definition	Tableau Formula	Business Value	Chart Type
Total Admissions	Count of unique patient hospital admissions in time frame.	COUNTD([Admission ID])	Baseline capacity demand measurement.	KPI Card / Sparkline
Occupancy Rate (%)	Percentage of total active operational beds occupied.	(SUM([Occupied Beds]) / SUM([Total Beds])) * 100	Informs admission triage & expansion planning.	Bullet Graph / Gauge Card
Average Length of Stay (ALOS)	Average days a patient occupies a hospital bed.	AVG(DATEDIFF('day', [Admission Date], [Discharge Date]))	Primary indicator of inpatient throughput efficiency.	Line Chart / Box Plot
30-Day Readmission Rate	% of patients readmitted within 30 days of discharge.	(COUNTD(IF [Days To Readmit] <= 30 THEN [Patient ID] END) / COUNTD([Patient ID])) * 100	Crucial indicator for care quality and discharge planning.	Donut Chart / Bar Chart
Bed Utilization Rate	Proportion of bed-hours occupied vs. available.	SUM([Occupied Bed Hours]) / SUM([Available Bed Hours])	Highlights asset turnarounds and capacity bottlenecks.	Heatmap / Stacked Bar
Department Efficiency Score	Weighted score based on turnaround times and throughput.	([Target Turnaround Time] / AVG([Actual Turnaround Time])) * 100	Evaluates clinical performance and resource alignment.	Horizontal Bar Chart

Implementation Note: All rate calculations utilize fixed Level of Detail (LOD) expressions, such as {FIXED [Department ID] : ...}, to prevent aggregations from distorting when global filters are applied.

7. HOSPITAL OVERVIEW WIREFRAME

High-level macro layout for executive leadership, emphasizing broad metric visibility.

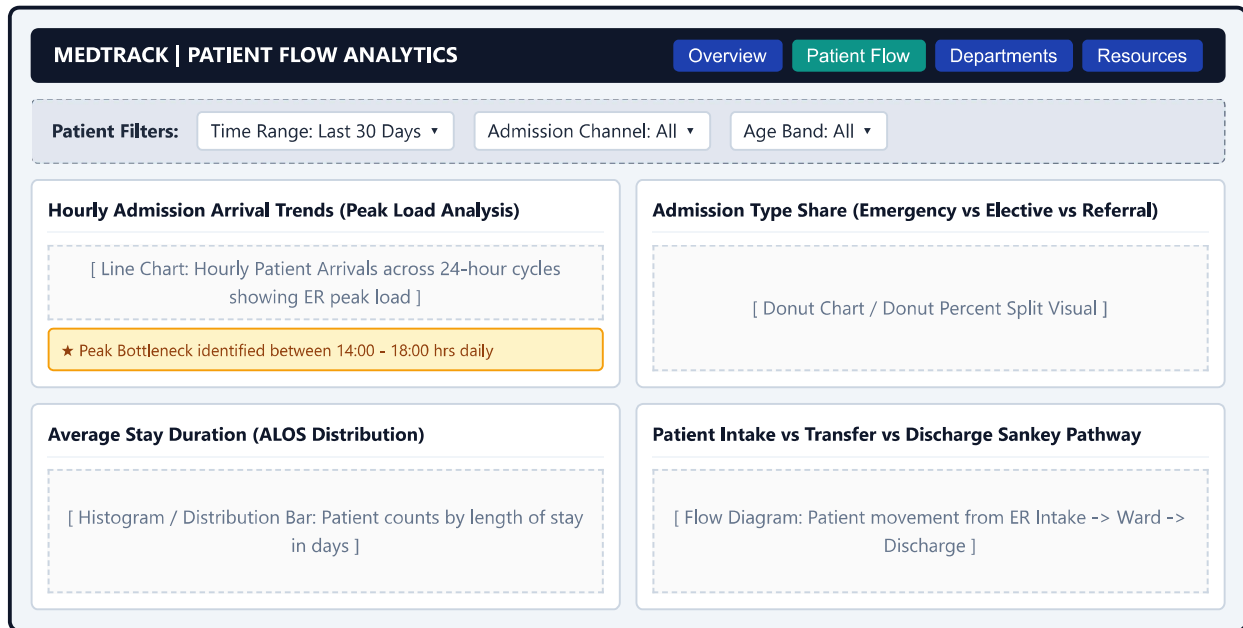


Component Breakdown

- **Top KPI Cards:** Provide immediate visibility into high-level hospital health metrics.
- **Admissions Trend Chart:** Dual-axis visual identifying historical growth and surge periods.
- **Occupancy & Readmission Breakdown:** Enables cross-departmental comparison to spot operational risk.

8. PATIENT FLOW WIREFRAME

Focuses on throughput bottlenecks, admission types, and discharge efficiency.



Interaction Design Features

- **Peak Load Identification:** Visual flags on peak hourly arrival times help manage shift coverage.
- **Sankey Movement Model:** Maps how patients move from initial triage to their final discharge ward.
- **Interactive Drill-Throughs:** Clicking an admission channel filters length-of-stay distributions in real time.

9. DEPARTMENT ANALYTICS WIREFRAME

Comparative analytics highlighting unit-level operational performance and treatment capacity.



Analytics Value & Usage

This view helps identify operational imbalances across clinical units. For instance, high patient volume alongside lower efficiency scores points to potential staffing or procedural bottlenecks in that department.

10. RESOURCE UTILIZATION WIREFRAME

Monitoring physical assets, bed availability, critical medical equipment, and staff coverage.



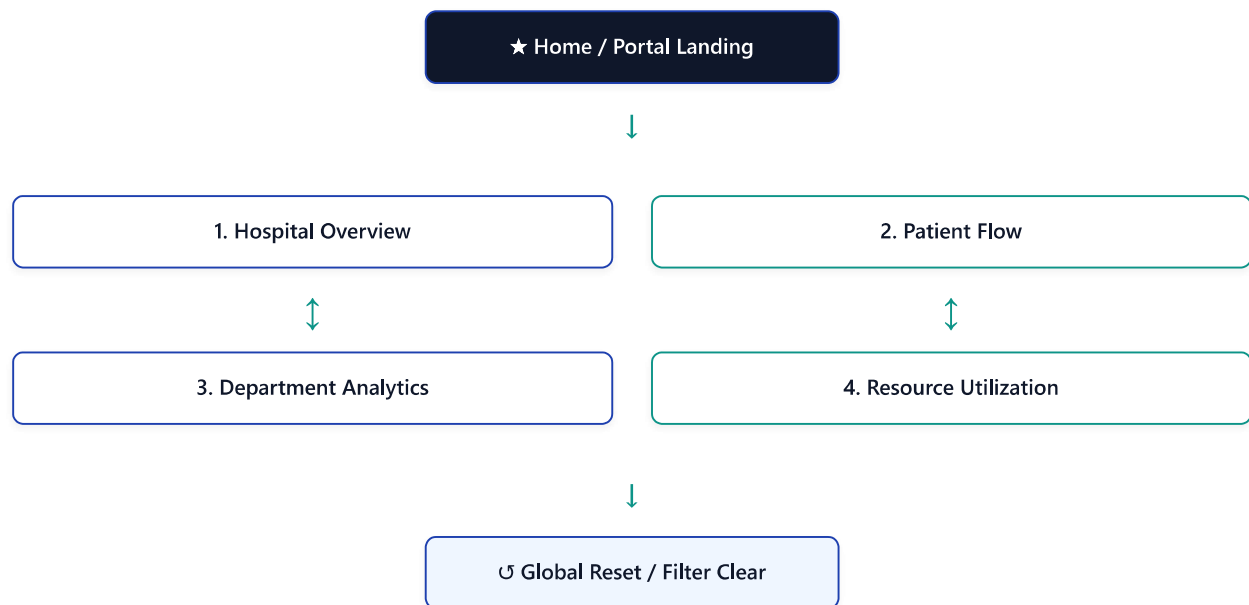
Operational Impact

- **Preventive Maintenance Tracking:** Flags heavily utilized equipment approaching maintenance limits.
- **Staffing Alignment:** Highlights shifts where nurse-to-patient ratios exceed safety targets.

11. INTERACTIVE NAVIGATION ARCHITECTURE

The MedTrack system uses a standardized top navigation bar on every dashboard page, giving users a seamless experience as they move between different levels of operational data.

11.1 Structural Navigation Flow Chart



11.2 Key Navigation Features

- **Tab Switchers:** Built using Tableau Parameter Actions for smooth, seamless view switching.
- **Contextual Drills:** Clicking a specific department in the *Overview* view filters the *Department Analytics* page accordingly.
- **Breadcrumb Trail:** Clear visual cues show users exactly where they are in the dashboard hierarchy.

12. FILTER & ACTION PLANNING

To ensure fast load times and intuitive usability, cross-dashboard interactivity is governed by a structured filter and action logic matrix.

Filter Category	Scope / Target	Mechanism	Functional Behavior
Global Date Range	All Dashboards	Relative Date Filter	Filters all calculations to YTD, QTD, or custom date ranges across all views.
Facility / Campus	All Dashboards	Data Source Filter	Filters operational data by selected physical hospital location.
Department Action	Overview → Department	Dashboard Filter Action	Clicking a department in the overview automatically opens and filters the department-specific view.
Hover Highlight	Single View Charts	Highlight Action	Hovering over a patient category highlights that cohort across all charts on the active page.
Reset Action	All Views	Parameter Reset Button	Clears all selected filters and resets dashboard parameters back to their default state.

Optimization Strategy: Apply-buttons are enabled on multi-select filters to prevent Tableau from executing unnecessary intermediate queries while users make their selections.

13. COLOR PALETTE & DESIGN STANDARDS

The MedTrack UI follows standard enterprise design principles, using an accessible, healthcare-focused color palette to maximize readability and reduce cognitive load.

13.1 Enterprise Color System

Color Element	Hex Code	Visual Sample	UI Application
Primary Navy	#0F172A		Headers, main titles, structural frames, dark accents.
Brand Royal Blue	#1E40AF		Primary data series, active navigation tabs, major KPIs.
Accent Teal	#0D9488		Positive trends, highlights, active selections, primary indicators.
Soft Blue Light	#EFF6FF		Alternate table row fills, card background fills.
Alert Amber/Red	#DC2626		Capacity warnings, threshold breaches, high readmissions.

13.2 Typography & Grid Layout

- **Font Family:** Segoe UI / Arial sans-serif (clean, universally supported fonts).
- **Hierarchy:** Dashboard Titles (18pt Bold), Section Headers (12pt Bold), KPI Values (20pt Bold), Data Labels (8pt Regular).
- **Grid Alignment:** Built on a strict 8px grid system to ensure consistent padding and clean alignment across all views.

14. PROJECT DELIVERABLES MATRIX

The complete deliverable package for Module 4 consists of technical documentation, prototype files, and interaction designs prepared for evaluation.

Deliverable Asset	Format / Type	Description / Purpose
dashboard_storyboard.pdf	PDF Document	Comprehensive Module 4 technical planning storyboard and wireframe specification (This document).
medtrack_prototype.twbx	Tableau Packaged Workbook	Functional Tableau prototype workbook containing interconnected dashboards and sample datasets.
Dashboard UI Mockups	Embedded Assets	Low and mid-fidelity wireframes detailing layouts for all four planned operational views.
Interactive Logic Plan	Technical Specification	Detailed plan for parameter actions, drill-through pathways, and cross-filtering behaviors.

Sign-off Readiness: This document completes the planning phase for **Module 4 – Dashboard Planning & Prototyping**. The specifications herein serve as the baseline blueprint for technical development in Modules 5 and 6.

15. FUTURE ENHANCEMENTS & ROADMAP

While the current scope fulfills core operational visualization needs, future phases will integrate advanced analytical and machine learning capabilities into the MedTrack ecosystem.

15.1 Technology Roadmap

- **Predictive Patient Admission Forecasting:** Integrating Time-Series forecasting models (ARIMA / Prophet) to predict ER patient surges up to 48 hours in advance.
- **Real-Time Data Streaming:** Connecting Tableau to live HL7/FHIR hospital streams via Kafka pipelines for sub-minute occupancy updates.
- **AI-Powered Discharge Optimization:** Using machine learning models to identify patients with a high risk of readmission before they are discharged.
- **Cloud Data Warehouse Integration:** Transitioning extracted data sources to a live Snowflake data warehouse for scalable performance.

15.2 Strategic Conclusion

This storyboard serves as the architectural foundation for the MedTrack Hospital Analytics suite. By defining key metrics, user roles, wireframe layouts, and interaction rules upfront, we ensure that technical development in **Module 5 (Dashboard Construction)** and **Module 6 (Testing & Deployment)** aligns directly with strategic business objectives.

Summary: The planned MedTrack dashboard system bridges the gap between operational healthcare data and executive decision-making—delivering an intuitive, high-performance visual analytics platform built on solid architecture principles.